TTAM USER GUIDE

This document is the user guide for the A**T**R **T**akeoff **A**nalysis **M**odule **TTAM** version H2/O9 OCT.2025 This TTAM version meets requirements of IATA / *SCAP Interface Specification - Takeoff* / 18th Edition August 2016.

H2: SCAP takeoff interface version number

O9: FOS module1 computation program version number

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1. Introduction

All airlines must provide performance data to enable their crews to operate aircraft safely, economically and in accordance with their applicable certification rules and operating practices. Basic data has in the past been provided by the aircraft manufacturer in various manuals (AFM, Operations Manual, Performance Engineer's Manual) in hard copy form. Progressively this has been supplemented by data provided digitally to facilitate data entry into programs developed either by the airlines or by aircraft manufacturers.

Most airlines use programs supplied by the aircraft manufacturers for each aircraft type, and interface these programs with their own computer operating environment, airfield database, and data output format requirements.

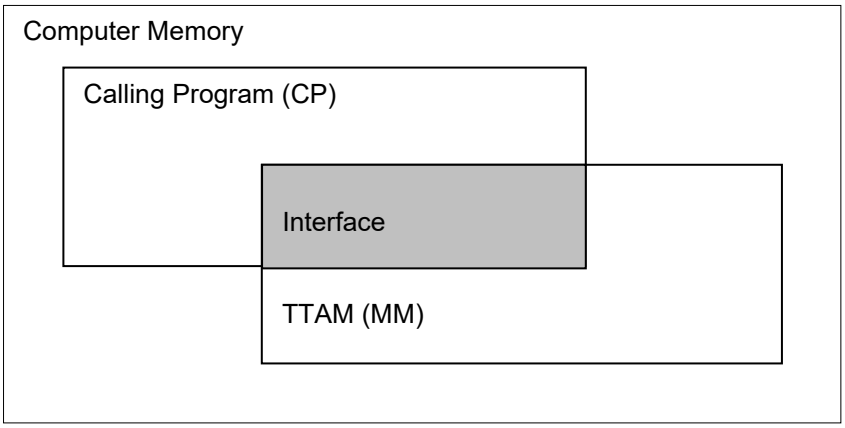
In order to reduce the development effort for the airlines (interfacing with different aircraft manufacturers' programs) and manufacturers (meeting individual airline requirements for customer specific features) there are clear advantages in standardizing as much as possible the input/output features of these programs, and also data formats.

Having recognized this potential benefit, IATA, ATA and several manufacturers set up a Standard Computerized Airplane Performance (SCAP) Working Group. The SCAP Working Group try to achieve maximum possible standardization of the takeoff performance program interface. The result of that effort is a Specification which defines the interfaces requirements for aircraft manufacturer supplied takeoff analysis software for implementation in the airline user environment.

The SCAP takeoff specification is applicable to civil, transport category, aircrafts and are published as joint IATA/ATA documents. Copies of the specification may be ordered at IATA.

2. General

The IATA specification describes the interface between a takeoff computation Manufacturer Module (MM) and an airline's calling program (CP). This interface is a part of the storage area of the computer memory that is shared by the MM and the CP as shown in the following figure.



The MM is provided by the aircraft manufacturer. It contains logic and data which allow to compute takeoff performance for a single point calculation, i.e. one temperature, one pressure altitude, one wind and one set of runway data. The MM is to be used as a single subroutine. The MM, called **TTAM** (A**T**R **T**akeoff **A**nalysis **M**odule), is available for all ATR42 and ATR72 models. The TTAM meets the requirements of the IATA takeoff specification. TTAM is an input/output interface which is plugged on top of the existing basic FOS Module 1 computation program.

TTAM meets the requirements of the IATA specification as regards the single point computation process philosophy and the interfaced input/output parameters. It contains no optional features conflicting with the specification but may contain some minor deviations which are clearly identified in this user guide.

The CP is implemented by the airline and it contains the procedure that suit the airline's needs. The CP is the main program.

Airline's CP and ATR's MM TTAM can transfer data to and from through the interface common area. CP and MM must be written in FORTRAN 77 programming language and must be compiled and linked together prior to execution.

The CP and the MM must contain exactly the same definition of COMMON BLOCKS and CHARACTER statements as described in the Input interface (Chapter 3) and the Output interface (Chapter 4) which means same COMMON labels, same parameter positions within the COMMON, same array dimensions and character lengths. Input and output data can be created using ASCII-based or XML standards.

The CP initialises the input parameters with actual values, or 9×10^{20} (for unspecified numerical parameters) or blanks (for unspecified alphanumerical parameters). Then, the CP calls the TTAM subroutine which process the input, reads the performance database, computes the case and fills the output interface with the output results or default values. The default values will be equal to 9×10^{20} for numerical parameters or blank for alphanumerical parameters.
Finally, the CP processes the output and ends.

3. Input data

The takeoff input interface consists of two FORTRAN COMMON BLOCKS: a numerical COMMON BLOCK labelled TINPN and an alphanumerical COMMON BLOCK labelled TINPA.

3.1. Numerical Common Block TINPN

The numerical COMMON BLOCK labelled TINPN contains 10 one-dimensional arrays of single precision REAL, each identified by a 4 character name. FORTRAN 77 conventions imply REAL*4 format for all data in this COMMON BLOCK.
The 4 character names of the arrays in TINPN are the following:

- POPT Program options
- CONF Aircraft configuration
- XMET Meteorological conditions
- RWYD Runway data
- OBSD Obstacle data
- FPTD Flight path turn data
- UNIT Units
- SPIA Supplemental performance interface A
- SPIB Supplemental performance interface B
- SPIC Supplemental performance interface C

Both CP and MM must contain the same following declaration statements:

```
  COMMON /TINPN/POPT(200),CONF(200),XMET(200),RWYD(200),OBSD(200),
 +FPTD(200),UNIT(200),SPIA(200),SPIB(200),SPIC(200)
```

3.1.1. POPT – Program options

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPT(1)	CALCULATION OPTION Allows the user to produce different levels of output.	0. = Full output 1. = Short output 2. = N/A Other =N/A	NONE	Flag 0. : From brake release to final takeoff (FTO). Flag 1. : From brake release to screen height (15ft or 35ft).
POPT(2)	V1/VR RANGE CHOICE	0. = Full AFM range 1. = N/A	NONE	
POPT(3)	V1/VR RANGE MINIMUM VALUE If POPT(2) = 1.	N/A		
POPT(4)	V1/VR RANGE MAXIMUM VALUE If POPT(2) = 1.	N/A		
POPT(5)	V1/VR CHOICE FLAG (According to AFM) Allows the user to select the mode of computation and to produce an output according to his choice.	0. = Optimum (Unbalanced). Output is Min V1 and Max V1 1. = Fixed V1/VR 2. = N/A 3. = N/A 4. = N/A 5. = N/A 6. = N/A 7. = N/A	NONE	When only one value of V1 is possible, results are in Min V1 and Max V1 positions in the output.

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPT(6)	V1/VR RATIO If POTP(5)=1., user will provide here the selected V1/VR. This ratio must conform to POPT(2) selection.	Selected V1/VR ratio.	NONE	This ratio is computed according to AFM presentation (i.e. with CAS).
POPT(7)	V2/VS CHOICE According to AFM presentation.	0. = Optimum V2/VS 1. = Fixed V2/VS	NONE	
POPT(8)	V2/VS RATIO If POPT(7)=1., user will provide here selected V2/VS ratio.	Selected V2/VS ratio	NONE	V2/VS ratio is computed with CAS.
POPT(9)	SCREEN HEIGHT AT END OF RUNWAY Height attained over the end of TOD.	15. = 15ft (or 4.57m) 35. = 35ft (or 10.67m)	UNIT(4)	15ft (or 4.57m): not applicable on dry runway. 35ft (or 10.67m): applicable on contaminated runway provided aircraft database contains relevant tables.
POPT(10)	OBSTACLE CLEARANCE Net flight path obstacle vertical clearance.	15. = 15ft (or 4.57m) 35. = 35ft (or 10.67m)	UNIT(4)	15ft (or 4.57m): not applicable on dry runway.
POPT(11)	TAKEOFF FLIGHT PATH Only within AFM limits. For time limit, check POPT(26)	0. = Second Segment Flight Path 1. = N/A 2. = N/A 3. = N/A 4. = N/A Other = N/A	NONE	
POPT(12)	ALIGNMENT ALLOWANCE FOR TORA AND TODA Line up distance. This distance will be subtracted from the beginning OF TORA and TODA.		UNIT(3)	
POPT(13)	ALIGNMENT ALLOWANCE FOR ASDA This distance will be subtracted from ASDA.		UNIT(3)	
POPT(14)	SURFACE CONDITION/TYPE OF CONTAMINATION Different types of runway contamination.	0. = Dry 1. = Wet (certified) 2. = Standing water 3. = Slush 4. = Compacted snow 5. = Dry snow 6. = Wet snow 7. = Ice 8. = N/A 9. = N/A 10. = N/A 11. = Selected MU or braking action value 12. = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. For valid values 1., 2., 3., 4., and 6., when certified data is computed the result should not be greater than the dry result (0. value).

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPT(15)	DEPTH OF CONTAMINATION Applicable only if POPT(14) = 2,3,5 or 6		UNIT(5)	
POPT(16)	SELECTED MU OR BRAKING ACTION VALUE OR RCR VALUE. May be applicable if POPT(14) = 5 or 6	Between 0 & 1 = Input MU value 1. = Poor 2. = Medium to poor 3. = Medium 4. = Medium to good 5. = Good 6. = Certification value	NONE	
POPT(17)	ENGINE STATUS FOR CONTAMINATED RUNWAYS	0. = One engine out concept 1. = N/A	NONE	
POPT(18)	FIXED GROSS LEVEL-OFF HEIGHT FLAG	0. = Output is Max and Min level-off heights. 1. = N/A 2. = N/A 3. = N/A	NONE	
POPT(19)	LOWEST LIMIT GROSS LEVEL-OFF HEIGHT		UNIT(4)	Actual lowest limit set in aircraft database is 400 ft.
POPT(20)	TRACE FLAG LEVEL Parameter current values are written to a file when trace is requested.	0. = No trace 1. = Trace		
POPT(21)	LOGICAL UNIT NUMBER FOR TRACE OUTPUT This output unit is used when POPT(20)≠0.	Any Logical Unit Number valid for output in user's installation.	NONE	A Logical Unit Number is an integer. Notice that POPT(21) is a REAL type variable.
POPT(22)	LOGICAL UNIT NUMBER FOR THE DATABASE Needed when the database must be read from a file.	The logical number of the file containing the Database.	NONE	A Logical Unit Number is an integer. Notice that POPT(22) is a REAL type variable.
POPT(23)	SELECTED MAXIMUM COMPUTATIONAL WEIGHT FLAG	0. = AFM performance chart limits 1. = Selected value	NONE	Output weights are limited by the AFM Performance Chart limits. CLIMIT(1) can be further limited by setting POPT(23)=1. and entering an appropriate value in POPT(24). Notice that the flight path and speeds are computed for the resulting limited takeoff weight (CLIMIT(1)).

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPT(24)	SELECTED MAXIMUM COMPUTATIONAL WEIGHT VALUE Only when POPT(23) = 1.		UNIT(8)	Output weights are limited by the AFM Performance Chart limits. CLIMIT(1) can be further limited by setting POPT(23)=1. and entering an appropriate value in POPT(24). Notice that the flight path and speeds are computed for the resulting limited takeoff weight (CLIMIT(1)).
POPT(25)	DERATED TAKEOFF THRUST LEVEL	N/A		
POPT(26)	TAKEOFF THRUST TIME LIMIT	N/A		Defined in aircraft database, which complies with regulation.
POPT(27)	V2/VS RANGE	0. = Full range 1. = N/A	NONE	The V2/VS value actually used is in CLIMIT(4) of output.
POPT(28)	V2/VS RANGE MINIMUM VALUE IF POPT(27)=1.	N/A		
POPT(29)	V2/VS RANGE MAXIMUM VALUE IF POPT(27)=1.	N/A		
POPT(30)	FORWARD CENTRE OF GRAVITY POSITION LIMIT FLAG	0. = Standard 1. = N/A 2. = N/A 3. = N/A Other = N/A	NONE	
POPT(31)	INDICATOR FOR PRECALCULATION USE	N/A		
POPT(32)	AIRPLANE CENTER OF GRAVITY	N/A		
POPT(33)	SECOND SEGMENT GRADIENT FLAG	0. = Normal (according to regulations) 1. = N/A	NONE	
POPT(34)	SECOND SEGMENT GRADIENT VALUE If POPT(33) = 1	N/A		
POPT(35)	DRAG INDEX	N/A		
POPT(36)	TIME HISTORY FLAG	0. = No time history Other = N/A	NONE	
POPT(37)	LOGICAL UNIT NUMBER FOR TIME HISTORY OUTPUT	N/A		
POPT(38)	OUTPUT DISTANCE ADJUSTEMENT FLAG	N/A		
POPT(39)	MINIMUM V1 RESTRICTION FLAG	0. = No restrictions 1. = N/A	NONE	
POPT(40)	MAXIMUM DIFFERENCE BETWEEN MINIMUM V1 AND VR	N/A		
POPT(41)	MAXIMUM STOPPING MARGIN FLAG	0. = No restriction 1. = N/A 2. = N/A	NONE	

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPT(42)	MAXIMUM STOPPING MARGIN VALUE USED IN THE COMPUTATION OF MINIMUM V1	N/A		
POPT(43)	THRUST CUTBACK CALCULATION FLAG	0. = No calculation 1. = N/A	NONE	
POPT(44)	THRUST CUTBACK ENGINE OUT GRADIENT VALUE If POPT(43)=1.	N/A		
POPT(45)	THRUST CUTBACK FLAPS POSITION If POPT(43)=1.	N/A		
POPT(46)	THRUST CUTBACK HEIGHT ABOVE RUNWAY If POPT(43)=1.	N/A		
POPT(47) thru POPT(48)	BLANK SPACES AVAILABLE			
POPT(49)	PERFORM DRY CHECK. Applicable to WET and CONTAMINATED runways.	0. = YES (DEFAULT) 1. = NO	NONE	
POPT(50) thru POPT(200)	BLANK SPACES AVAILABLE			

3.1.2. CONF – Aircraft configuration

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONF(1)	SLAT/FLAP CHOICE	0. = Optimum (highest weight) 1. = Selected configuration 2. = N/A 3. = N/A 4. = Selected configuration (alphanumerical value). Refer to FDCFGI in common TINPA.	NONE	If CONF(1) = 1. let FDCFGI blank. If (CONF(1) = 4. let CONF(2) and CONF(3) unassigned.
CONF(2)	SLAT POSITION	0.	Degrees	No slats on ATR.
CONF(3)	FLAP POSITION	Refer to AFM for valid flap positions	Degrees	E.G.: 15.
CONF(4)	AIR CONDITIONING	0. = N/A 1. = On 2. = N/A 2. = Off 3. = N/A 4. = N/A	NONE	1. only available for 42-500 and all 72.
CONF(5)	ANTI-ICING	1. = Off 2. = N/A 2. = N/A 3. = N/A 4. = N/A 5. = N/A 6. = Atmospheric icing condition Other = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONF(6)	GROUND SPOILERS	0. = Armed 1. = N/A 2. = Inoperative 3. = N/A 4. = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONF(7)	ANTISKID	0. = On 1. = N/A 2. = Totally off	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONF(8)	THRUST REVERSE CREDIT	0. = No reverse thrust credit 1. = Credit for all thrust reversers. 2. = N/A 3. = N/A 4. = N/A	NONE	Selecting CONF(8)=1. means that all thrust reversers are operative before takeoff. Selecting CONF(8)=1 for dry runway leads to fatal error.
CONF(9)	LANDING GEAR POSITION AFTER FIRST SEGMENT	0. = Normal 1. = Extended 2. = N/A Others = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. Value 1. cannot be combined with atmospheric icing condition CONF(5) = 6.

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONF(10)	MAIN FUSELAGE LANDING GEAR POSITION BEFORE TAKEOFF	N/A		
CONF(11)	APU.	N/A		
CONF(12)	POWER MANAGEMENT SYSTEM	0. = Normal 1. = Autofeather inoperative 2. = E.C.U off 3. = one S.C.U inoperative 4. = one torque indicator inoperative 5. = E.C.U on - ATPCS off 6. = E.E.C off 7. = A.F.U off 8. = ATPCS off 9. = one ACW generator inoperative 10. = Uptrim function inoperative	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONF(13)	AUTO BRAKES	N/A		
CONF(14)	TIRE PRESSURE	0. = Normal tire 1. = Low pressure tire	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONF(15)	TIRE PROFILE	0. = Normal 1. = N/A	NONE	
CONF(16)	TIRE SPEED	N/A		Actual maximum limit is set in aircraft database
CONF(17)	BRAKE ENERGY CHOICE	0. = Normal 1. = N/A	NONE	
CONF(18)	BRAKE ENERGY VALUE	N/A		
CONF(19)	SPEED BRAKES	N/A		
CONF(20)	BRAKES CONFIGURATION	0. = All brakes operating 1. = One brake inoperative 2. = N/A 3. = N/A Other = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONF(21)	TAKEOFF THRUST AT BRAKE RELEASE	N/A		
CONF(22)	ENGINE FAILURE WARNING SYSTEM FLAG	N/A		
CONF(23)	HYDRAULIC PUMP INOPERARTIVE	N/A		
CONF(24)	ONE ENGINE OUT FERRY FLIGHT	N/A		
CONF(25)	SPARE ENGINE CARRIAGE	N/A		
CONF(26)	VORTEX DISSIPATORS	N/A		
CONF(27)	WHEEL TACHOMETER(S)	N/A		
CONF(28)	FLAP POSITION When CONF(1) = 0., 2 or 3. then the minimum flap position to consider is provided here.	N/A		

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONF(29)	FLAP POSITION When CONF(1) = 0., 2 or 3. then the maximum flap position to consider is provided here.	N/A		
CONF(30)	THRUST REVERSE LEVEL SELECTION	N/A		
CONF(31)	LANDING LIGHTS AUTO RETRACTION	N/A		
CONF(32)	FLAPS FAULT / LOCKED	0. = Normal 1. = Retracted	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. Value 1. cannot be combined with atmospheric icing condition CONF(5) = 6.
CONF(33)	SLATS FAULT / LOCKED	N/A		
CONF(34)	THRUST REVERSER DELAY FLAG	N/A		
CONF(35)	ADDITIONAL THRUST REVERSER TIME DELAY	N/A		
CONF(36)	APPROACH FLAP POSITION	N/A		
CONF(37)	APPROACH AIR CONDITIONING	N/A		
CONF(38)	APPROACH ANTI-ICING	N/A		
CONF(39)	NOSE-WHEEL STEERING	N/A		
CONF(40)	PROP RPM FOR NOISE ABATMENT FLAG	N/A		
CONF(41)	PROP RPM VALUE IF CONF(40) = 1.	N/A		
CONF(42)	PROP AUTO FEATHER	N/A		
CONF(43)	IN FLIGHT ICING	N/A		
CONF(44)	CENTER GEAR BRAKES CONFIGURATION	N/A		
CONF(45)	GROUND IDLE	N/A		
CONF(46)	FUEL TANK PUMP	N/A		
CONF(47)	FAN AIR VALVE	N/A		
CONF(48)	EXTERNAL INSTALLATIONS	N/A		
CONF(49)	ENGINE INTERMIX	N/A		
CONF(50)	APU DOOR POSITIONS	N/A		
CONF(51)	SEMI-LEVERED GEAR	N/A		
CONF(52)	APPROACH THRUST POWER FLAG	N/A		
CONF(53)	TAILSKID CONFIGURATION	N/A		

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONF(54)	DE-ICING / ANTI-ICING FLUID	0. = None 1. = Method 1 2. = Method 2	NONE	Only valid for type II/IV fluids Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. Value 1.
CONF(55)	TAKEOFF THRUST POWER	0. = Normal Take Off 1. = Reserve Take Off (RTO) at brake release	NONE	1. only valid for 42-500 and 72-500 with PW127M/XT engines
CONF(56)	TAKEOFF RATING	0. = Optimum (highest weight) 1. = Boost off 2. = Boost on 3. = Super boost on	NONE	0. Optimizes both CONF(55) and CONF(56) Values 1. to 3. only valid with multi boost file
CONF(57) thru CONF(200)	BLANK SPACES AVAILABLE			

3.1.3. XMET – Meteorological conditions

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
XMET(1)	TEMPERATURE		UNIT(1)	This parameter can be used for certified actual temperature or for assumed temperature. [Refer to CLIMIT(34)]. If UNIT(1) = 2. or 3., XMET(1) = ISA Deviation Temperature according to pressure altitude.
XMET(2)	WIND COMPONENT Unfactored wind component along the runway	+ (Positive) = Headwind - (Negative) = Tailwind	UNIT(9)	Data calculation is based on wind component measured at 10m.
XMET(3)	CROSSWIND COMPONENT	N/A		
XMET(4)	ALTITUDE FLAG User choice to measure airport altitude	0. = Airport pressure altitude 1. = QNH 2. = N/A	NONE	
XMET(5)	ALTITUDE VALUE Altitude for which computation will be performed, expressed in terms of XMET(4).	Set here Altitude pressure altitude, in accordance with XMET(4)	UNIT(6)	Units must be in accordance with choice in XMET(4).
XMET(6)	ACTUAL OUTSIDE AIR TEMPERATURE (OAT)	N/A		
XMET(7)	TEMPERATURE INVERSION FLAG	N/A		
XMET(8)	TEMPERATURE AT INVERSION ALTITUDE	N/A		
XMET(9)	INVERSION ALTITUDE	N/A		
XMET(10) thru XMET(200)	BLANK SPACES AVAILABLE			

3.1.4. RWYD – Runway data

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
RWYD(1)	ELEVATION Airport geometrical elevation with respect to Sea Level. For information only if XMET(4) = 0. (Airport Reference Elevation).		UNIT(4)	
RWYD(2)	FLAG FOR RUNWAY LENGTH DATA TYPE Next three parameters will relate to this flag.	0. = TORA, ASDA, TODA. 1. = TORA, stopway length, clearway length.	NONE	
RWYD(3)	TORA (RUNWAY LENGTH)		UNIT(3)	
RWYD(4)	ASDA OR STOPWAY LENGTH		UNIT(3)	ASDA = TORA + stopway length.
RWYD(5)	TODA OR CLEARWAY LENGTH		UNIT(3)	TODA = TORA + clearway length.
RWYD(6)	SLOPE FOR TODA This slope is used to compute TOD and effect on the flight path.	+ (Positive) = Up-hill - (Negative) = Down-hill	%	
RWYD(7)	SLOPE FOR ASDA This slope is used to compute ASD.	N/A		
RWYD(8)	RUNWAY SURFACE	0. = Normal 1. = Gravel 2. = Grooved and/or Porous 3. = Dry unpaved with CBR less than 15 4. = Dry unpaved with CBR equal or greater than 15 5. = Grass runway with CBR less than 15 6. = Grass runway with CBR equal or greater than 15 7. = Coral 8. = N/A	NONE	Provided aircraft database contains relevant tables. Check which surfaces are available in FOS Module 1 interface.
RWYD(9)	ASDA MULTIPLICATION FACTOR This parameter allows user to multiply ASDA by a correction factor. This parameter must always be filled.	0.<K< 1.	NONE	Manufacturers and operators use conservative ASDA values for several purposes. This factor is applied after subtracting alignment allowance, POPT(13). ASDA for computation = RWYD(9)*(RWYD(4) - POPT(13)) + RWYD(10)
RWYD(10)	ASDA INCREMENT VALUE This parameter allows to apply an increment (decrement if negative) to actual ASDA.		UNIT(3)	

Array Position	**Parameter**	**Valid Values** (N/A means Not Available)	**Units**	**Comments**
RWYD(11)	RUNWAY WIDTH		UNIT(3)	This parameter may have an influence on Vmcg value.
RWYD(12) thru RWYD(200)	BLANK SPACES AVAILABLE			

3.1.5. OBSD – Obstacles

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
OBSD(1)	NUMBER OF OBSTACLES	0 to 20 obstacles.	NONE	
OBSD(2)	OBSTACLE DATUM FLAG Obstacle distance and geometric height origin flag.	0. = Beginning of TORA 1. = End of TORA	NONE	
OBSD(3)	LONGITUDINAL DISTANCE OF OBSTACLE N°1 According to OBSD(2).		UNIT(3)	Obstacles can be arbitrarily ordered. Limiting obstacle identifier in output interface will correspond to input order.
OBSD(4)	GEOMETRICAL HEIGHT OF OBSTACLE N°1 According to OBSD(2).		UNIT(4)	
OBSD(5)	LATERAL DISTANCE OF OBSTACLE N°1 From runway centreline (+ right, - left). Only useful if FPTD(1) not equal to 0.	N/A	UNIT(3)	
OBSD(60)	LONGITUDINAL DISTANCE OF OBSTACLE N°20		UNIT(3)	
OBSD(61)	GEOMETRICAL HEIGHT OF OBSTACLE N°20		UNIT(4)	
OBSD(62)	LATERAL DISTANCE OF OBSTACLE N°20	N/A	UNIT(3)	
OBSD(63) thru OBSD(200)	BLANK SPACES AVAILABLE			

3.1.6. FPTD – Flight path data

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
FPTD(1)	FLIGHT PATH FLAG User options of type of turn	0. = No turn 1. = N/A 2. = N/A		
FPTD(2)	TURN RADIUS OR BANK ANGLE.	N/A		
FPTD(3)	START OF TURN FLAG.	N/A		
FPTD(4)	DISTANCE OR HEIGHT VALUE.	N/A		
FPTD(5)	END OF TURN FLAG.	N/A		
FPTD(6)	HEADING CHANGE OR DISTANCE.	N/A		
FPTD(7)	FLIGHT PATH WIND VELOCITY.	N/A		
FPTD(8)	FLIGHT PATH WIND DIRECTION.	N/A		
FPTD(9)	OBSTACLE SELECTION FLAG IN CASE OF TURN.	N/A		
FPTD(10)	FLIGHT PATH SPLAY ANGLE.	N/A		
FPTD(11)	FLIGHT PATH LATERAL HALFWIDTH AT THE END OF ONE ENGINE OUT TOD.	N/A		
FPTD(12)	FLIGHT PATH LATERAL HALFWIDTH AT THE END OF FLIGHT PATH SPLAY.	N/A		
FPTD(13)	FLAG FOR START OF FLIGHT PATH.	N/A		
FPTD(14)	DISTANCE BETWEEN THE END OF THE RUNWAY AND THE INITIAL FLIGHT PATH LAT. H/WIDTH.	N/A		
FPTD(15)	MIN. GROSS WEIGHT FOR CONSIDERATION IN A TURN.	N/A		
FPTD(16)	MAXIMUM BANK ANGLE FOR RADIUS TURNS.	N/A		
FPTD(17) thru FPTD(200)	BLANK SPACES AVAILABLE			

3.1.7. **UNIT – Units**

Array Position	**Parameter**	**Valid Values** (N/A means Not Available)	**Units**	**Comments**
UNIT(1)	TEMPERATURE	0. = Celsius 1. = Fahrenheit 2. = ISA deviation in Celsius 3. = ISA deviation in Fahrenheit		
UNIT(2)	TIME	N/A		
UNIT(3)	LENGTH AND HORIZONTAL DISTANCES	0. = Meters 1. = Feet		
UNIT(4)	HEIGHT, ELEVATION, PRESSURE ALTITUDE	0. = Meters 1. = Feet		Height for obstacles. Elevation and Pressure Altitude for Airport and Runway.
UNIT(5)	DEPTH OF CONTAMINATION	0. = N/A 1. = Inches		
UNIT(6)	PRESSURE (QFE,QNH)	0. = HectoPascal 1. = Inches Hg.		(mb)
UNIT(7)	TIRE PRESSURE	N/A		
UNIT(8)	WEIGHT	0. = Kilograms 1. = Pounds 2. = N/A		
UNIT(9)	SPEED (GROUND AND AIR)	0. = N/A 1. = N/A 2. = Knots 3. = N/A		
UNIT(10) thru UNIT(200)	BLANK SPACES AVAILABLE			
SLOPE AND GRADIENT ALWAYS IN %				
ANGLES (BANK ANGLE, HEADING AND HEADING CHANGE) ALWAYS IN DEGREES				

3.1.8. SPIA – Supplemental performance interface

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
SPIA(1)	MULTIPLICATION FACTOR(K) ON CALCULATED 1-ENGINE OUT ASD		NONE	K*CALCULATED ASD + A
SPIA(2)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 1-ENGINE OUT ASD		UNIT(3)	
SPIA(3)	MULTIPLICATION FACTOR(K) ON CALCULATED 1-ENGINE OUT TOD		NONE	K*CALCULATED TOD + A
SPIA(4)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 1-ENGINE OUT TOD		UNIT(3)	
SPIA(5)	MULTIPLICATION FACTOR(K) ON CALCULATED 1-ENGINE OUT TOR		NONE	K*CALCULATED TOR + A
SPIA(6)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 1-ENGINE OUT TOR		UNIT(3)	
SPIA(7)	MULTIPLICATION FACTOR(K) ON CALCULATED ALL-ENGINE OUT ASD	N/A	NONE	K*CALCULATED ASD + A
SPIA(8)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED ALL-ENGINE OUT ASD	N/A	UNIT(3)	
SPIA(9)	MULTIPLICATION FACTOR(K) ON CALCULATED ALL-ENGINE OUT TOD		NONE	K*CALCULATED TOD + A
SPIA(10)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED ALL-ENGINE OUT TOD		UNIT(3)	
SPIA(11)	MULTIPLICATION FACTOR(K) ON CALCULATED ALL-ENGINE OUT TOR		NONE	K*CALCULATED TOR + A
SPIA(12)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED ALL-ENGINE OUT TOR		UNIT(3)	
SPIA(13)	MULTIPLICATION FACTOR(K) ON CALCULATED BALANCED FIELD LENGTH	N/A	NONE	K*CALCULATED FIELD LENGTH + A

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
SPIA(14)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED BALANCED FIELD LENGTH	N/A	UNIT(3)	K*CALCULATED FIELD LENGTH + A
SPIA(15)	MULTIPLICATION FACTOR(K) ON CALCULATED 1st SEGMENT WEIGHT		NONE	K*CALCULATED 1st SEGMENT WEIGHT + A
SPIA(16)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 1st SEGMENT WEIGHT		UNIT(8)	
SPIA(17)	MULTIPLICATION FACTOR(K) ON CALCULATED 2nd SEGMENT WEIGHT		NONE	K*CALCULATED 2nd SEGMENT WEIGHT + A
SPIA(18)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 2nd SEGMENT WEIGHT		UNIT(8)	
SPIA(19)	MULTIPLICATION FACTOR(K) ON CALCULATED FINAL TAKEOFF WEIGHT		NONE	K*CALCULATED FINAL TAKEOFF WEIGHT + A
SPIA(20)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED FINAL TAKEOFF WEIGHT		UNIT(8)	
SPIA(21)	MULTIPLICATION FACTOR(K) ON CALCULATED 1st SEGMENT GRADIENT	N/A	NONE	K*CALCULATED 1st SEGMENT GRADIENT + A
SPIA(22)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 1st SEGMENT GRADIENT	N/A	%	
SPIA(23)	MULTIPLICATION FACTOR(K) ON CALCULATED 2nd SEGMENT GRADIENT		NONE	K*CALCULATED 2nd SEGMENT GRADIENT + A
SPIA(24)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED 2nd SEGMENT GRADIENT		%	
SPIA(25)	MULTIPLICATION FACTOR(K) ON CALCULATED FINAL TAKEOFF GRADIENT	N/A	NONE	K*CALCULATED FINAL TAKEOFF GRADIENT + A
SPIA(26)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED FINAL TAKEOFF GRADIENT	N/A	%	
SPIA(27)	MULTIPLICATION FACTOR(K) ON CALCULATED V1 LIMITED BY BRAKE ENERGY		NONE	K*CALCULATED V1MBE + A

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
SPIA(28)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED V1 LIMITED BY BRAKE ENERGY		UNIT(9)	K*CALCULATED V1MBE + A
SPIA(29)	MULTIPLICATION FACTOR(K) ON CALCULATED V1 LIMITED BY VMCG		NONE	K*V1 LIMITED BY VMCG + A
SPIA(30)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED V1 LIMITED BY VMCG		UNIT(9)	
SPIA(31)	MULTIPLICATION FACTOR(K) ON CALCULATED VMCA		NONE	K* CALCULATED VMCA + A
SPIA(32)	INCREMENT/DECREMENT VALUE(A) ON CALCULATED VMCA		UNIT(9)	
SPIA(33) thru SPIA(200)	BLANK SPACES AVAILABLE			

3.1.9. SPIB – Supplemental performance interface

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
SPIB(1) thru SPIB(200)	BLANK SPACES AVAILABLE			

3.1.10. SPIC – Supplemental performance interface

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
SPIC(1) thru SPIC(200)	BLANK SPACES AVAILABLE			

3.2. Alphanumerical Common Block TINPA

The alphanumerical COMMON BLOCK labelled TINPA contains 13 parameters and one array of three elements for comments. Every one of these parameters has an independent number of characters and has to be left justified with blanks filled.

Both CP and MM must contain the same following declaration statements:

```
      CHARACTER*20 ACTYPE,APTIDT,BRAKES,CERTIF,DATE,DBOPTI,ENGTYP,
     +             FDCFGI,RATING,REVNBR,RWYNBR
      CHARACTER*255 ACMDBI,MISCLA
      CHARACTER*132 COMENT(20)
      COMMON /TINPA/ACTYPE,ENGTYP,RATING,BRAKES,CERTIF,REVNBR,DATE,
     +       DBOPTI,ACMDBI,APTIDT,RWYNBR,COMENT,MISCLA,FDCFGI
```

Array Position	**Parameter**	**Valid Values** (N/A means Not Available)	**Format**	**Comments**
ACTYPE	AIRCRAFT TYPE.	N/A	CHARACTER*20	SEE ACMDBI array
ENGTYP	ENGINE TYPE.	N/A	CHARACTER*20	
RATING	RATING LEVEL.	N/A	CHARACTER*20	
BRAKES	BRAKES TYPE.	N/A	CHARACTER*20	
CERTIF	CERTIFICATION BASIS.	Blank for mono regulation file For multi regulation file: JARA for JAR regulation FAR for FAR regulation	CHARACTER*20	Provided aircraft database contains relevant tables. Check which regulations are available in FOS Module 1 interface.
REVNBR	REVISION NUMBER.	N/A	CHARACTER*20	SEE ACMDBI array
DATE	REVISION DATE.	N/A	CHARACTER*20	
DBOPTI	DATABASE INPUT MODIFICATION OPTIONS.	N/A	CHARACTER*20	
ACMDBI	MANUFACTURER DATABASE IDENTIFICATION CODE. Set of characters, manufacturer defined, that identifies a specific database.	Performance data file name.	CHARACTER*255	Enter here the logical name corresponding to the specified aircraft.
APTIDT	AIRPORT IDENTIFICATION.		CHARACTER*20	
RWYNBR	RUNWAY IDENTIFICATION.		CHARACTER*20	
COMENT(1) thru COMENT(20)	COMMENTS		CHARACTER*132	Allows writing here engine failure procedure or other comments.
MISCLA	MISCELANEOUS. Provision for future input data.	N/A	CHARACTER*255	
FDCFGI	SLAT/FLAP CONFIGURATION		CHARACTER*20	E.G.: S00F15.

4. Output data

The IATA specification do not request manufacturers to provide all specified output parameters but as many parameters as feasible. Output parameters which are not available from the TTAM are identified by the code N/A in the COMMENTS column and are set to 9 x 10^{20}. Output parameters exceeding the limits are set to 9999999 and a message is displayed. The units described in chapter 3.1.7 apply to input parameters as well as to the expected output parameters.

The takeoff output interface consists of two COMMON BLOCKS : a numerical COMMON BLOCK labelled TOUTN and an alphanumerical COMMON BLOCK labelled TOUTA.

4.1. Numerical Common Block TOUTN

The numerical COMMON BLOCK labelled TOUTN contains 9 one-dimensional arrays of single precision REAL, each identified by a 6 character name. FORTRAN 77 conventions imply REAL*4 format for all data in this COMMON BLOCK.
The 6 character names of the arrays in TOUTN are the following:

- CLIMIT Limiting condition
- RWYSEG Runway segment
- FIRSEG First segment climb
- SECSEG Second segment climb
- ACCSEG Acceleration segment
- FINSEG Final segment climb
- OBSLIM Obstacle clearance
- TRNDAT Turn data
- SPOUTA Supplemental performance output

Both CP and MM must contain the same following declaration statement:

```
 COMMON /TOUTN/CLIMIT,RWYSEG,FIRSEG,SECSEG,ACCSEG,FINSEG,OBSLIM,
+      TRNDAT,SPOUTA
```

4.1.1. CLIMIT – Limiting condition

Array Position	**Parameter**	**Units**	**Comments** (N/A means Not Available)
CLIMIT(1)	Maximum takeoff weight.	UNIT(8)	This weight is the lowest of the limiting weights.
CLIMIT(2)	Slat position.	NONE	
CLIMIT(3)	Flap position.	NONE	
CLIMIT(4)	V2/VS ratio associated with the maximum takeoff weight CLIMIT(1).	NONE	V2/VS ratio calculated with speeds in CAS.
CLIMIT(5)	V2	UNIT(9)	Takeoff safety speed in IAS.
CLIMIT(6)	VR	UNIT(9)	Rotation speed in IAS.
CLIMIT(7)	Minimum V1 if POPT(5) = 0 (optimum V1/VR) or V1 corresponding to the fixed V1/VR ratio selected if POPT(5) = 1.	UNIT(9)	Decision speed in IAS. If there is only one V1, all the associated values [CLIMIT(8) to CLIMIT(12)] will be put in the minimum V1 and maximum V1 blocks.
CLIMIT(8)	The first limitation code associated with the CLIMIT(7) speed.	NONE	
CLIMIT(9)	The second limitation code associated with the CLIMIT(7) speed.	NONE	
CLIMIT(10)	The required takeoff run associated with the CLIMIT(7) speed and the maximum takeoff weight CLIMIT(1).	UNIT(3)	

Array Position	Parameter	Units	Comments (N/A means Not Available)
CLIMIT(11)	The required takeoff distance associated with the CLIMIT(7) speed and the maximum takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(12)	The required accelerated-stop distance associated with the CLIMIT(7) speed and the maximum takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(13)	Maximum V1 if POPT(5)=0. (optimum V1/VR) or V1 corresponding to the fixed V1/VR ratio selected if POPT(5) = 1	UNIT(9)	Decision speed in IAS.
CLIMIT(14)	The first limitation code associated with the CLIMIT(13) speed.	NONE	
CLIMIT(15)	The second limitation code associated with the CLIMIT(13) speed.	NONE	
CLIMIT(16)	The required takeoff run associated with the CLIMIT(13) speed and the maximum takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(17)	The required takeoff distance associated with the CLIMIT(13) speed and the maximum takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(18)	The required accelerate-stop distance associated with the CLIMIT(13) speed and the maximum takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(19)	The balanced V1.		N/A
CLIMIT(20)	The first limitation code associated with the balanced V1 CLIMIT(19).		N/A
CLIMIT(21)	The second limitation code associated with the balanced V1 CLIMIT(19)		N/A
CLIMIT(22)	The required takeoff run associated with the balanced V1 CLIMIT(19) and the maximum takeoff weight CLIMIT(1).		N/A
CLIMIT(23)	The required takeoff distance associated with the balanced V1 CLIMIT(19) and the maximum takeoff weight CLIMIT(1).		N/A
CLIMIT(24)	The required accelerate-stop distance associated with balanced V1 CLIMIT(19) and maximum takeoff weight CLIMIT(1).		N/A
CLIMIT(25)	The all engine takeoff run associated with the max takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(26)	The all engine takeoff distance associated with max takeoff weight CLIMIT(1).	UNIT(3)	
CLIMIT(27)	Primary takeoff thrust setting parameter flag for standard rating.		N/A
CLIMIT(28)	Takeoff thrust setting value according to CLIMIT(27) for standard rating.		N/A
CLIMIT(29)	Secondary takeoff thrust setting parameter flag for standard rating.		N/A
CLIMIT(30)	Takeoff thrust setting value according to the output CLIMIT(29) for standard rating.		N/A
CLIMIT(31)	V1 GO on contaminated runway.		N/A
CLIMIT(32)	V1 STOP on contaminated runway.		N/A
CLIMIT(33)	The speed limited by maximum brake energy (VMBE) and associated with the maximum takeoff weight CLIMIT(1).	UNIT(9)	
CLIMIT(34)	Temperature envelop flag.		N/A
CLIMIT(35)	The minimum V1/VR ratio associated with the maximum takeoff weight CLIMIT(1).		Computed with CAS.
CLIMIT(36)	The maximum V1/VR ratio associated with the maximum takeoff weight CLIMIT(1).		Computed with CAS.
CLIMIT(37)	The balanced V1/VR ratio associated with the maximum takeoff weight CLIMIT(1).		N/A

Array Position	Parameter	Units	Comments (N/A means Not Available)
CLIMIT(38)	The takeoff weight limited by fuel dump exemption.		N/A
CLIMIT(39)	Primary takeoff thrust setting parameter flag for reserve rating.		N/A
CLIMIT(40)	Takeoff thrust setting value according to CLIMIT(39).		N/A
CLIMIT(41)	Secondary takeoff thrust setting flag for reserve rating.		N/A
CLIMIT(42)	Takeoff thrust setting according to CLIMIT(41).		N/A
CLIMIT(43)	Maximum structural takeoff weight.	UNIT(8)	N/A
CLIMIT(44)	Calculated maximum usable assumed temperature.	UNIT(1)	N/A
CLIMIT(45)	Maximum assumed temperature.	UNIT(1)	N/A
CLIMIT(46)	Takeoff thrust break temperature.	UNIT(1)	N/A
CLIMIT(47)	Limitation code for the minimum V1 accelerate-stop distance CLIMIT(12).	NONE	N/A
CLIMIT(48)	Limitation code for the maximum V1 accelerate-stop distance CLIMIT(18).	NONE	N/A
CLIMIT(49)	Limitation code for the balanced accelerate-stop distance CLIMIT(24).	NONE	N/A
CLIMIT(50)	Cutback thrust setting value according to CLIMIT(39).	NONE	N/A
CLIMIT(51)	Blank spaces available		
CLIMIT(52)	RUNWAY DRY CHECK CODE	NONE	Valid if POPT(14) <> 0. 0 or 9E20 = Check Not Done 1. = CLIMIT(1) is limited by runway condition POPT(14) 2. = CLIMIT(1) is limited by DRY runway performance
CLIMIT(53) thru CLIMIT(54)	Blank spaces available		
CLIMIT(55)	TAKEOFF THRUST POWER	NONE	0. = Normal Take Off 1. = Reserve Take Off (RTO) at brake release
CLIMIT(56)	TAKEOFF RATING	NONE	1. = Boost off 2. = Boost on 3. = Super boost on
CLIMIT(57) thru CLIMIT(200)	Blank spaces available		

4.1.2. RWYSEG – Runway segment

Array Position	Parameter	Units	Comments (N/A means Not Available)
RWYSEG(1)	The brake energy weight associated with the CLIMIT(7) speed.	UNIT(8)	
RWYSEG(2)	The brake energy weight associated with the CLIMIT(13) speed.	UNIT(8)	
RWYSEG(3)	The brake energy weight associated with the balanced V1 CLIMIT(19).		N/A
RWYSEG(4)	The speed limited by minimum control on ground (VMCG) or minimum certified V1).	UNIT(9)	In IAS
RWYSEG(5)	The weight limited by tire speed.	UNIT(8)	
RWYSEG(6)	The maximum allowable runway limited weight.	UNIT(8)	
RWYSEG(7)	Unassigned		N/A
RWYSEG(8)	Unassigned		N/A
RWYSEG(9)	Unassigned		N/A
RWYSEG(10)	The speed limited by VMU.		In IAS
RWYSEG(11)	The speed limited by minimum VR.		N/A
RWYSEG(12)	The speed limited by minimum V2.		In IAS
RWYSEG(13)	The speed limited by 1.05Vmca.	UNIT(9)	In IAS
RWYSEG(14)	Field length limitation code.		N/A
RWYSEG(15)	V1 associated with RWYSEG(6).		N/A
RWYSEG(16)	Minimum V1 associated with RWYSEG(6).		N/A
RWYSEG(17)	Maximum V1 associated with RWYSEG(6).		N/A
RWYSEG(18)	VR associated with RWYSEG(6).		N/A
RWYSEG(19)	V2 associated with RWYSEG(6).		N/A
RWYSEG(20)	Field length maximum weight further limited by obstacle clearance.		N/A
RWYSEG(21)	V1 associated with RWYSEG(20).		N/A
RWYSEG(22)	V1/VR ratio associated with RWYSEG(6).	NONE	
RWYSEG(23)	V1 Min/VR ratio associated with RWYSEG(6).		N/A
RWYSEG(24)	V1 Max/VR ratio associated with RWYSEG(6).		N/A
RWYSEG(25)	Field length limited takeoff weight (pure).	UNIT(8)	
RWYSEG(26)	Distance to “reference zero” point above brakes release with reference to RWYSEG(7).	UNIT(3)	N/A
RWYSEG(27)	Height of “reference zero” point above brakes release with reference to RWYSEG(7).	UNIT(4)	N/A
RWYSEG(28)	Engine thrust time limit flag	NONE	N/A
RWYSEG(29)	All engine field length limit weight.	UNIT(8)	
RWYSEG(30) thru RWYSEG(200)	Blank spaces available		

4.1.3. FIRSEG – First segment climb

Array Position	Parameter	Units	Comments (N/A means Not Available)
FIRSEG(1)	The weight limited by the first segment condition.	UNIT(8)	
FIRSEG(2)	The distance between the end of the takeoff distance and the end of the first segment for the maximum takeoff weight CLIMIT(1).	UNIT(3)	
FIRSEG(3)	The net height at the end of the first segment for the maximum takeoff weight CLIMIT(1).	UNIT(4)	
FIRSEG(4)	The gross height at the end of the first segment for the maximum takeoff weight CLIMIT(1).	UNIT(4)	
FIRSEG(5) thru FIRSEG(200)	Blank spaces available		

4.1.4. SECSEG – Second segment climb

Array Position	Parameter	Units	Comments (N/A means Not Available)
SECSEG(1)	The weight limited by the second segment condition.	UNIT(8)	
SECSEG(2)	The net second segment climb gradient associated with the maximum takeoff weight CLIMIT(1).		N/A
SECSEG(3)	The gross second segment climb gradient associated with the maximum takeoff weight CLIMIT(1).	%	
SECSEG(4)	The horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of the second segment at the minimum level-off height or at the fixed level-off height.	UNIT(3)	If the fixed gross level-off height flag [input POPT(18)] is equal to 0., the minimum level-off height is the value selected in the input POPT(19) or can be higher if there are obstacles. If the minimum and the maximum heights are equal, all the data will be put only in the maximum height block. Deviation to SCAP specification.
SECSEG(5)	The horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of the second segment at the maximum level-off height.	UNIT(3)	
SECSEG(6)	The net height at the end of the extended second segment for the maximum takeoff weight CLIMIT(1).		N/A
SECSEG(7)	The gross height at the end of the extended second segment for the maximum takeoff weight CLIMIT(1).		N/A
SECSEG(8)	The gross height at the end of the extended second segment converted into pressure altitude relative to airport.		N/A
SECSEG(9)	The horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of the extended second segment.		N/A
SECSEG(10)	The one engine out target pitch attitude.		N/A
SECSEG(11)	The all engine target pitch attitude.		N/A
SECSEG(12) thru SECSEG(200)	Blank spaces available		

4.1.5. ACCSEG – Acceleration segment

Array Position	Parameter	Units	Comments (N/A means Not Available)
ACCSEG(1)	The net height of the minimum level-off for the maximum takeoff weight CLIMIT(1).	UNIT(4)	
ACCSEG(2)	The gross height of the minimum level-off for the max takeoff weight CLIMIT(1).	UNIT(4)	
ACCSEG(3)	The gross height of the minimum level off converted to pressure altitude relative to airport pressure altitude.	UNIT(4)	ACCSEG(3) = ACCSEG(2) converted to pressure altitude.
ACCSEG(4)	The net flight path horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of acceleration segment made at the minimum level-off height or at the fixed level-off height.	UNIT(3)	N/A
ACCSEG(5)	The gross flight path horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of the acceleration segment made at the minimum level-off height or at the fixed level off height.	UNIT(3)	N/A
ACCSEG(6)	The net height of the maximum level-off for the maximum takeoff weight CLIMIT(1).	UNIT(4)	
ACCSEG(7)	The gross height of the max level-off for the maximum takeoff weight CLIMIT(1).	UNIT(4)	
ACCSEG(8)	The gross height of the maximum level off converted pressure altitude relative to the airport pressure altitude.	UNIT(4)	ACCSEG(8)=ACCSEG(7) converted to pressure altitude.
ACCSEG(9)	The net flight path horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of the acceleration segment made at the maximum level-off height.	UNIT(3)	
ACCSEG(10)	The gross flight path horizontal distance between the end of the takeoff dist. required for the maximum takeoff weight CLIMIT(1) and the end of the acceleration segment made at the maximum level-off height.	UNIT(3)	N/A
ACCSEG(11) thru ACCSEG(200)	Blank spaces available		

4.1.6. FINSEG – Final segment climb

Array Position	Parameter	Units	Comments (N/A means Not Available)
FINSEG(1)	The weight limited by the final takeoff condition.	UNIT(8)	
FINSEG(2)	The net final takeoff climb gradient associated with the maximum takeoff weight CLIMIT(1).		N/A
FINSEG(3)	The gross final takeoff climb gradient associated with the maximum takeoff weight CLIMIT(1).	%	
FINSEG(4)	The horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1) and the end of the final segment corresponding to the acceleration segment made at the minimum level-off height.		N/A
FINSEG(5)	The net height corresponding to the FINSEG(4) distance.		N/A
FINSEG(6)	The gross height corresponding to the FINSEG(4) distance.		N/A
FINSEG(7)	The gross height corresponding to the FINSEG(4) distance converted to pressure altitude relative to airport pressure altitude.		N/A
FINSEG(8)	The horizontal distance between the end of the takeoff distance required for the max takeoff weight CLIMIT(1) and the end of final segment corresponding to the acceleration segment made at the maximum level-off height.		N/A
FINSEG(9)	The net height corresponding to the FINSEG(8) distance.		N/A
FINSEG(10)	The gross height corresponding to the FINSEG(8) distance.		N/A
FINSEG(11)	The gross height corresponding to the FINSEG(8) distance converted to pressure altitude relative to airport pressure altitude.		N/A
FINSEG(12)	The net horizontal distance between the end of the takeoff distance required for the maximum takeoff weight (CLIMIT(1)) and the end of the 5th segment corresponding to the acceleration segment at the minimum level-off height or at the fixed level-off height.		N/A
FINSEG(13)	The gross horizontal distance between the end of the takeoff distance required for the maximum takeoff weight (CLIMIT(1)) and the end of the 5th segment corresponding to the acceleration segment at the minimum level-off height or at the fixed level-off height.		N/A
FINSEG(14)	The net horizontal distance between the end of the takeoff distance required for the maximum takeoff weight (CLIMIT(1)) and the end of the 5th segment corresponding to the acceleration segment at the maximum level-off height or at the fixed level-off height.		N/A
FINSEG(15)	The gross horizontal distance between the end of the takeoff distance required for the maximum takeoff weight CLIMIT(1)) and the end of the 5th segment corresponding to the acceleration segment at the maximum level-off height or at the fixed level-off height.		N/A
FINSEG(16)	Blank spaces.		
FINSEG(16) thru FINSEG(200)	Blank spaces available		

4.1.7. OBSLIM – Obstacle clearance

Array Position	Parameter	Units	Comments (N/A means Not Available)
OBSLIM(1)	The weight limited by obstacle clearance associated with the minimum V1 or the V1 corresponding to the fixed ratio V1/VR selected if POPT(5)=1.	UNIT(8)	The weight limited by obstacle clearance associated with the fixed ratio (V1/VR) selected if POPT(5)=1. See NOTE(1).
OBSLIM(2)	The limiting obstacle number associated with the OBSLIM(1) weight and the minimum V1 or the V1 corresponding to the fixed ratio V1/VR selected if POPT(5)=1.	NONE	The limiting obstacle number associated with OBSLIM(1) weight and the fixed ratio (V1/VR) selected if POPT(5)=1. See NOTE(1).
OBSLIM(3)	The weight limited by obstacle clearance associated with the maximum V1.	UNIT(8)	The weight limited by obstacle clearance associated with the maximum V1/VR ratio if POPT(5)=0. See NOTE(2).
OBSLIM(4)	The limiting obstacle number associated with the OBSLIM(3) weight and the maximum V1.	NONE	The limiting obstacle number associated with the OBSLIM(3) weight and the maximum V1/VR ratio if POPT(5)=0. See NOTE(2).
OBSLIM(5)	The weight limited by obstacle clearance associated with the balanced V1.	UNIT(8)	N/A
OBSLIM(6)	The limiting obstacle number associated with the OBSLIM(5) weight and the balanced V1.	NONE	N/A
OBSLIM(7) thru OBSLIM(200)	Blank spaces available		

NOTE(1) : If OBSLIM(2) ≠ 0 and OBSLIM(1) ≠ CLIMIT(1) : The value given in array OBSLIM(1) is the weight limited by obstacle clearance calculated for the available runway lengths (brake energy not taken into account).

If OBSLIM(2) = 0 and OBSLIM(1) is defined : The value given in array OBSLIM(1) corresponds to a virtual obstacle limiting weight limited by a flight path minimum gradient 2.4%.

If OBSLIM(2) = 0 and OBSLIM(1) = 9999999 : The virtual obstacle limiting weight corresponding to the flight path minimum gradient is higher than the runway limited weight.

NOTE(2) : If OBSLIM(4) ≠ 0 and OBSLIM(3) ≠ CLIMIT(1) : The value given in array OBSLIM(3) is the weight limited by obstacle clearance calculated for the available runway lengths (brake energy and VMCG limitations not taken into account).

If OBSLIM(4) = 0 and OBSLIM(3) is defined : The value given in array OBSLIM(3) corresponds to a virtual obstacle limiting weight limited by a flight path minimum gradient of 2.4%.

If OBSLIM(4) = 0 and OBSLIM(3) = 9999999 : The virtual obstacle limiting weight corresponding to the flight path minimum gradient is higher than the runway limited weight.

4.1.8. TRNDAT – Turn data

Array Position	**Parameter**	**Units**	**Comments** (N/A means Not Available)
TRNDAT(1)	The bank angle at start of turn.		N/A
TRNDAT(2)	The height at start of turn.		N/A
TRNDAT(3)	The horizontal distance from end of runway to start of turn.		N/A
TRNDAT(4)	The radius at start of turn.		N/A
TRNDAT(5)	The bank angle at end of turn.		N/A
TRNDAT(6)	The height at end of turn.		N/A
TRNDAT(7)	The horizontal projected distance at end of turn from end of runway along the flight track.		N/A
TRNDAT(8)	The radius at end of turn.		N/A
TRNDAT(9)	The height at the start of the turn converted to pressure altitude above beginning of TORA.		N/A
TRNDAT(10)	The height at the end of the turn converted to pressure altitude above beginning of TORA.		N/A
TRNDAT(11) thru TRNDAT(200)	Blank spaces available		

4.1.9. SPOUTA – Supplemental performance output

Array Position	**Parameter**	**Units**	**Comments** (N/A means Not Available)
SPOUTA(1)	V flap retract.	UNIT(9)	N/A
SPOUTA(2)	V slat retract.	UNIT(9)	N/A
SPOUTA(3)	V min maneuver.	UNIT(9)	N/A
SPOUTA(4)	Vs	UNIT(9)	N/A
SPOUTA(5)	Vref	UNIT(9)	N/A
SPOUTA(6) thru SPOUTA(100)	Blank spaces available		

4.2. Alphanumerical Common Block TOUTA

The alphanumerical COMMON BLOCK labelled TOUTA contains one parameter for error code, three parameters of information, one array of ten elements for error messages and one array of three elements for miscellaneous information. Every one of these parameters has an independent number of characters and has to be left justified with blanks filled.

Both CP and MM must contain the same following declaration statements:

```
CHARACTER*20 ERFLAG
CHARACTER*20 MMDATE,MMIDNT
CHARACTER*255 ACMDBO
CHARACTER*80 ERRMSG(20),OUTMIS(20)
COMMON /TOUTA/ERFLAG,ERRMSG,MMIDNT,MMDATE,ACMDBO,OUTMIS
```

Array Position	**Parameter**	**Format**	**Comments**
ERFLAG	Error code flag	CHARACTER*20	A = Successful B = Input error C = Computation error D = TTAM has restricted input value to AFM performances chart(s) limits E = Performance restrictions do not allow computation of valid takeoff weight or speeds F = Successful. Warning: input value(s) not used for this A/C model. See trace file.
ERRMSG(1) thru ERRMSG(20)	Error messages	CHARACTER*80	See also trace output file (logical unit defined by POPT(21)) for more details on error and warning messages
MMIDNT	Manufacturer Module Identification	CHARACTER*20	Read here : TTAM
MMDATE	Manufacturer Module Date	CHARACTER*20	
ACMDBO	Aircraft Manufacturer database identification code.	CHARACTER*60	Read here same as in input array ACMDBI
OUTMIS(1) thru OUTMIS(20)	Miscellaneous - Provision for future output data	CHARACTER*80	

ANNEX

1. List of the limitation codes used

Code	Definition	Comments (N/U means Not Used by TTAM)
1	The structural or max. selected computational weight.	Corresponds to FOS module 1 code 1
2	The all engine takeoff distance limitation.	Corresponds to FOS module 1 code 7
3	The one engine out takeoff distance limitation.	Corresponds to FOS module 1 code 3
4	The accelerate stop distance limitation (all engine or one engine out).	Corresponds to FOS module 1 code 3
5	The one engine out accelerate stop distance limitation.	N/U
6	The all engine accelerate stop distance limitation.	N/U
7	The all engine acceleration limitation.	N/U
8	The one engine out acceleration limitation.	N/U
9	The minimum certified V1 (or VMCG) limitation.	Corresponds to FOS module 1 code 9
10	The tire speed limitation.	Corresponds to FOS module 1 code 5
11	The tire pressure limitation.	N/U
12	The maximum brake energy limitation.	Corresponds to FOS module 1 code 6
13	The minimum certified V2 (VMCA).	Corresponds to FOS module 1 code 9
14	The first segment climb gradient limitation.	Corresponds to FOS module 1 code 2
15	The second segment climb gradient limitation.	Corresponds to FOS module 1 code 2 It may indicate also 1st segment limitation.
16	The takeoff thrust time-limited limitation.	N/U
17	The acceleration segment (third segment) limitation.	Corresponds to FOS module 1 code 8
18	The final segment climb gradient limitation.	Corresponds to FOS module 1 code 8
19	The obstacle clearance limitation.	Corresponds to FOS module 1 code 4
20	The one engine out takeoff run limitation.	Corresponds to FOS module 1 code 3
21	The all engine takeoff run limitation.	Corresponds to FOS module 1 code 7
22	The minimum V1/VR limitation.	Corresponds to FOS module 1 code 3
23	The maximum V1/VR limitation.	Corresponds to FOS module 1 code 3
24	The uphill runway slope limitation.	Corresponds to FOS module 1 code 3
25	The one engine out runway length limitation.	Corresponds to FOS module 1 code 3
26	The all engine runway length limitation.	Corresponds to FOS module 1 code 7
27 to 50	(Provision)	N/U
51	Fuel dump exemption.	N/U

2. Error messages / Examples

The following messages are displayed in the logical unit assigned in input array POPT(21).

ERFLAG	Message	Reasons
B	INPUT EROR : PERFORMANCE DATA FILE NOT FOUND	Check file assigned in POPT(22).
B	INPUT ERROR:[....] NOT VALID	Value not defined by specification. E.G.:INPUT ERROR :POPT(1)=3. NOT VALID
B	INPUT ERROR:[....] NOT APPLICABLE FOR THIS A/C MODEL	Value not valid for the TTAM. E.G.:INPUT ERROR :POPT(1)=2. NOT APPLICABLE FOR THIS A/C MODEL
B	INPUT ERROR:CONFLICT BETWEEN [....] AND [....].	Conflict between two input variables.
B	INPUT ERROR:TOO MANY EFFECTS ARE USED (7 MAXIMUM).	The computation program FOS module 1 limits to 7 the number of 'non-basic' conditions selectable at a time.
B	INPUT ERROR:BAD OBST.LOCATION FOR OBST N° [..].	Longitudinal obstacle distance shorter than TODA.
F	***WARNING***: [....] NOT USED FOR THIS A/C MODEL.	Input parameter ignored by the computation program FOS module 1 - 'non-fatal' error. These parameters are noted 'N/A' on the input data list.
C	THE TABLE - ....- DOES NOT EXIST IN THE MODULE	
C	ILLEGAL GOTO IN	
C	BAD CONVERGENCE FOR VMCA	
C	BAD CONVERGENCE FOR TEMP=.... WIND=.... V2/VS=.... CONT=.... ERROR=...	
C	BAD CONVERGENCE FOR VMCG	
C	INCORRECT CONVERGENCE IN V1FIX FOR CONT.:.....	
C	NUMBER OF ABSC. NOT ENOUGH DETECTED BY *FINTV* - SEE CURVE BELOW ...	
C	NUMBER OF ABSC. NOT ENOUGH IN TABLE N°....	
C	ARG. GRATER THAN 99999999.	
C	ABSCISSAS NOT INCREASING	
C	ABSCISSAS NOT ENDED BY 9999999.	
C	ILLEGAL GOTO IN JNT FOR NTAB(....)	
C	DIMENSION PROBLEM FOR NTAB(....)	
C	BLOCK VNUM TOO SHORT - MINIMUM SIZE TO USE :	
C	PROVIDED NUMBER OF RECORDS IN MODULE .... IS FALSE	Performance data set may have been corrupted - Contact ATR Training centre
C	VMCA OPTIMIZATION NOT POSSIBLE FOR FIXED V2/VS OPTION	Check input data.
C	COMPUTATION ERROR *** MTOW LESS THAN PMIN	Check input data : resulting MTOW lower than minimum computational weight.
C	REVERSE EFFECT UNDEFINED IN PERFORMANCE DATA FILE	
C	THERE IS NO MODULE NAMED ..... IN THE MAIN FILE	
C	STOP 10 INPUT ERROR SEE GENERAL OUTPUT	This message comes in addition to other message in assigned Logical Unit.
C	LIMITATION NO: .... NOT USED	For information only.

C	T.O. DATA ON: .... ARE NOT AVAILABLE OR INCOMPLETE	Check input data and/or Performance data file in use.
C	T.O. DATA FOR RWY STATUS CODE (NIGC) = ... ARE NOT AVAILABLE	NIGC code is equivalent to POPT(14)+(15) and related to the surface condition/type and depth of contaminant - See also FOS module 1 and check the status of the performance data file.
E	VMCA LIMITATION - TAKEOFF NOT ALLOWED FOR: TEMPERATURE = ... WIND (OR QNH) = ...	
E	VMCG LIMITATION - TAKEOFF NOT ALLOWED FOR: TEMPERATURE = ... WIND (OR QNH) = ...	
	THE LATTER MESSAGES ARE RELATED TO: AIRPORT : ... RUNWAY: .... AIRCRAFT:	Message displayed each time there is an ERFLAG B,C,D,E, or F